12. The Rings

A ring breaks, grinds and returns: the source of the material, its dynamics, and the marks it leaves on moons and on the parent planet

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Abstract

Paper 11 established that a ring does not build moons. This paper develops that principle into a full mechanism for rings. It sets out where the material comes from, how it moves, what it does to the moons that pass through it, and what traces it leaves on the parent planet.

The paper makes four claims:

1. The material is original. A ring is made of the planet's own material. This material leaks from within near the corotation line and is trapped inside the Roche limit; it is not space dust. Saturn's rings are 95%–99% water ice and carry about 1.54×10^{19} kg, all in a thin disc in the equatorial plane. Dust arriving from every direction cannot form such a structure.

2. A ring breaks, grinds and returns; it does not build. Most of the material goes back into the planet, and only a little is pushed outward. At Saturn the flow has been measured directly: between 432 and 45,000 kg of material falls from the rings into the planet every second. At that rate the rings would vanish within 11 million to 1.1 billion years. They therefore form a steady flow fed from within. Over 4.5 billion years the required supply is 4 to 415 ring masses, but only one part in ten million to one hundred thousand of Saturn's mass. A ring outside the Roche limit, such as Quaoar's, does not gather into a moon either.

3. A ring's position is set relative to the corotation line. The rings of the four giant planets lie below or around the corotation line, at 0.5–1.4 corotation radii, where material falls inward. The rings of the three small bodies that have them (Haumea, Chariklo, Quaoar) lie above the line, at 1.25–2.2. Most of these are close to the 1:3 spin–orbit resonance at 2.08. When the corotation line migrates outward, the ring zone becomes a zone of infall and the ring disappears.

4. A ring marks the moon. A moon that passed through a ring carries its mark, and the kind of material sets the mark. An ice ring adds material to the moon, as in the equatorial belts of Pan, Atlas and Daphnis; a ring of silicate material grinds the moon. Deimos stayed long in the heart of the Martian ring and was ground all over. Phobos left through its lower edge and was ground only slightly. The mark also depends on the ring's thickness relative to the moon's diameter, on the crossing speed and on the moon's gravity.

The paper also summarises the signatures a ring leaves on its parent planet: measured at Saturn and required at Mars. It sets out the Sun's part in breaking up rings. Requirements, predictions and refutation criteria are collected in Part L.

`What this paper proposes`

- **Claim:** a planetary ring is a stage in a flow of material out of the planet and back into it. The material leaks from within, is trapped, breaks and is ground down, and most of it returns as ring rain. The ring exists as long as it is fed and as long as the corotation line lies near it. It does not build moons, but it changes the moons that pass through it: it circularises their orbits, pushes them outward, and leaves a mark on their surfaces according to the kind of material.
- **Key quantities:** the corotation line, $a_c = \left(GM/\Omega^2 \right)^{1/3}$; the Roche limit for ice, $a_R \approx 2.46\, R\, (\rho/\rho_{ice})^{1/3}$; Saturn's ring rain, 432–45,000 kg s⁻¹; the mass of Saturn's rings, 1.54×10^{19} kg.
- **What is required and what would refute it:** Part L.

Working environment

Quantity	Value	Source
Mass of Saturn's rings	1.54×10^{19} kg (about 40% of Mimas)	Iess et al. 2019
Composition of the A and B rings	95%–99% water ice	Cuzzi et al. 2010
Thickness of the dense rings	about 10–30 m; E ring: thousands of km	Cuzzi et al. 2010
Neutral ring rain, into the equator	4,800–45,000 kg s⁻¹	Waite et al. 2018

Ionised ring rain, along field lines	432–2,870 kg s^{-1}	O'Donoghue et al. 2019
Corotation line, in equatorial radii	Jupiter 2.24; Saturn 1.85; Uranus 3.24; Neptune 3.37	Calculation (A.3)
Hollows on Deimos and Phobos (median depth to width)	1.9% and 5.5%	Ernst et al. 2023; Paper 10

PART A Introduction: two lines around every planet

A.1 The question

All four giant planets have rings, yet the rings are very different:

- Saturn's are massive and bright.
- Jupiter's are thin dust.
- Uranus's and Neptune's are narrow and dark.

Three small bodies at the edge of the Solar System have dense rings, one of them far outside the Roche limit. The inner planets have no rings today, but Mars had one in the past [Paper 10; Ćuk et al. 2020]. No ring has been found around any moon.

Paper 11 defined the corotation line as the line around which moons are born, and established that a ring is a site of grinding, not of building. This paper asks how the same line, together with the Roche limit, sets several things: where a ring will be, what it is made of, how long it lives, and what it does to the bodies that pass through it.

A.2 The corotation line and the Roche limit

- **The corotation line:** the distance at which a body orbiting the planet completes one orbit in exactly one rotation of the planet, $a_c = \left(GM/\Omega^2 \right)^{1/3}$. Below the line, material orbits faster than the planet turns. The tide it raises on the planet lags behind it and brakes it, and it moves inward. Above the line the tide leads it and pushes it outward [Paper 11, A.6].
- **The Roche limit:** the distance inside which the planet's tidal force exceeds the self-gravity of a soft body, so that no such body can gather. For ice particles, $a_R \approx 2.46\, R\, (\rho/\rho_{ice})^{1/3}$, where ρ is the planet's density.

A.3 Where the rings lie

The table gives the position of the rings and of the Roche limit in units of the corotation radius. This puts planets of different sizes and spin rates on the same scale.

Body	Corotation line (km)	Rings (km)	Rings in units of a_c	Roche limit for ice, in a_c	Position relative to the line
Jupiter	160,000	92,000–226,000 (halo, main, gossamer)	0.57–1.41	1.25	Around the line
Saturn	111,600	66,900–136,800 (D to A)	0.60–1.23	1.21	Around the line; outer edge at the Roche limit
Uranus	82,700	41,800–51,100	0.51–0.62	0.85	Below the line
Neptune	83,500	41,900–62,900	0.50–0.75	0.89	Below the line
Haumea	1,104	2,287	2.07	2.27	Above the line
Chariklo	186	391; 405	2.10; 2.18	1.71	Above the line
Quaoar	2,020	2,520; 4,057	1.25; 2.01	0.83	Above the line; both outside the Roche limit

The numbers show two families:

- **The giant planets:** their rings lie below or around the corotation line, inside the Roche limit or at its edge, in the zone where material falls into the planet. At Saturn, the outer edge of the A ring (136,800 km) lies almost exactly on the Roche limit for ice (135,300 km).
- **The small bodies:** their rings lie above the line, mostly close to 2.08 corotation radii. This is the distance at which a particle completes one orbit while the body turns three times, the 1:3 resonance [Sicardy et al. 2019].

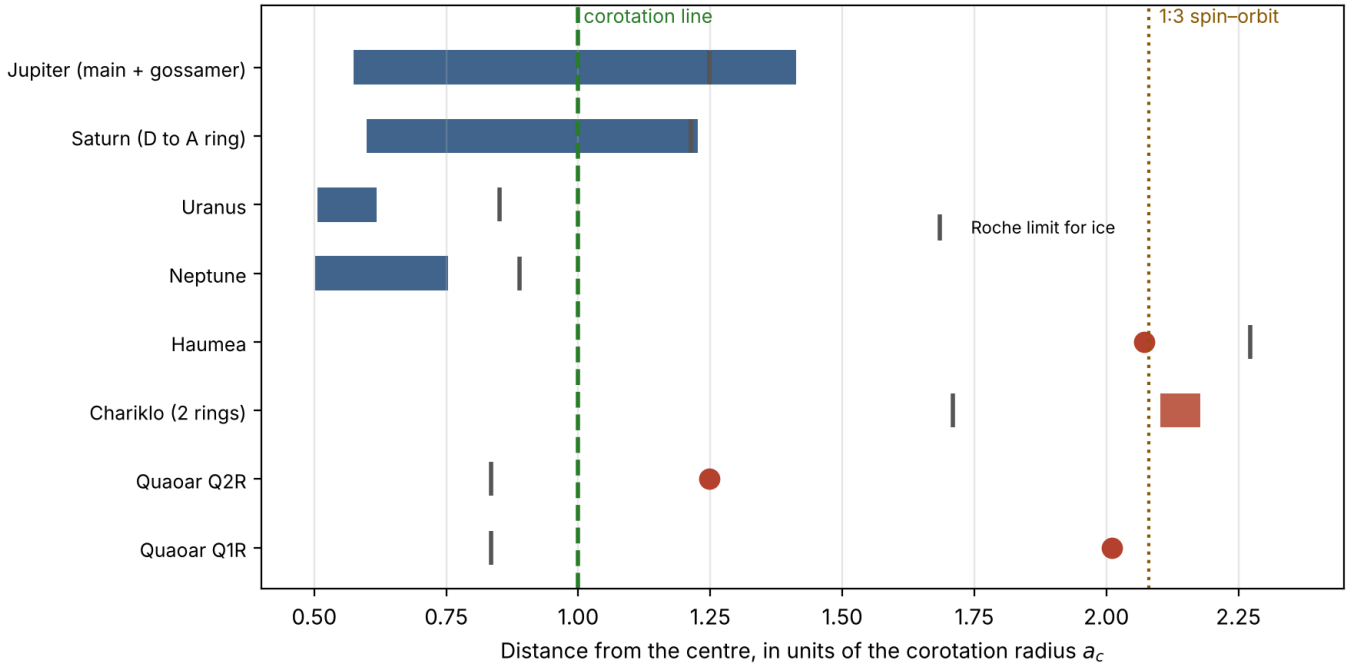
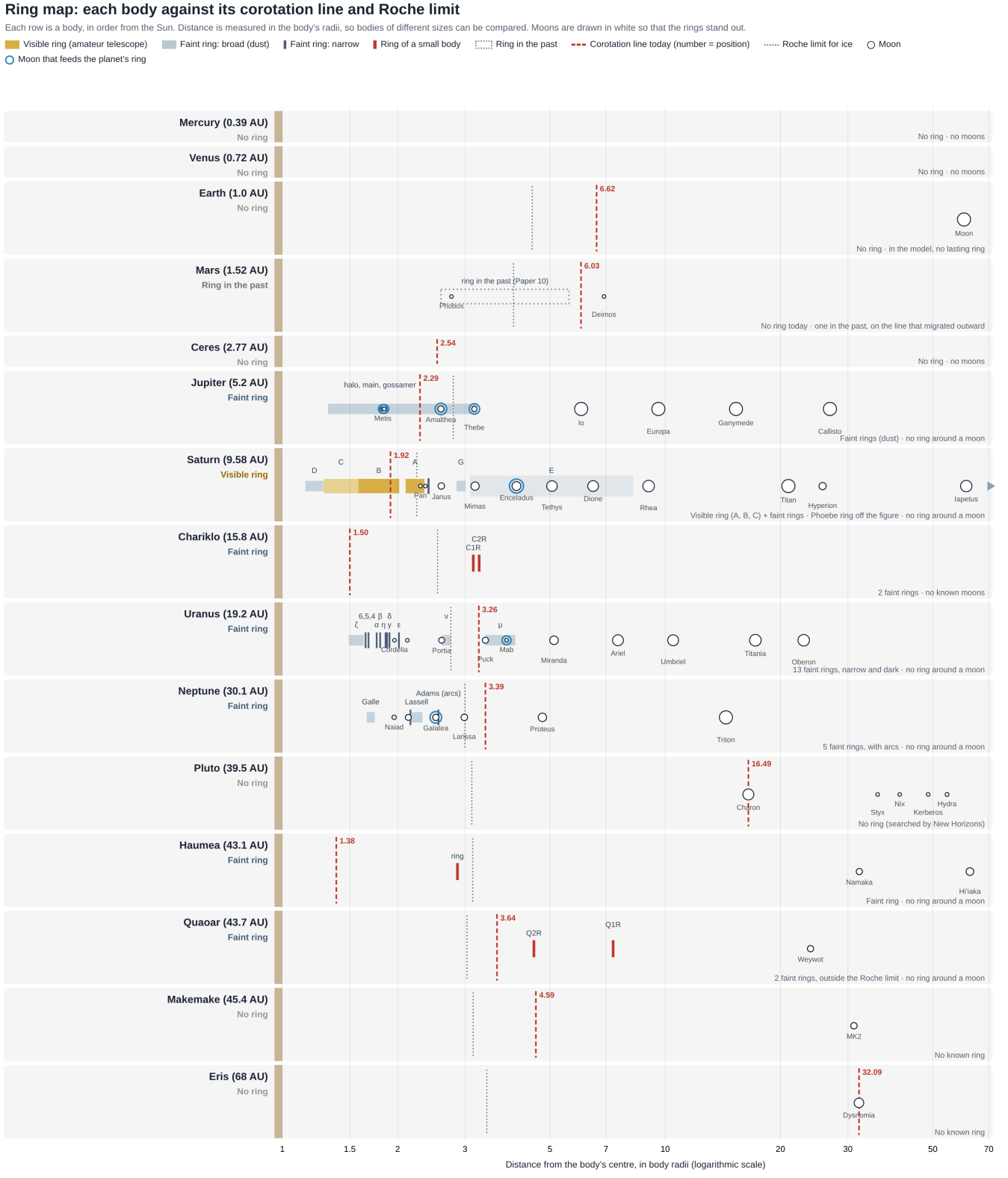


Figure A.1. Position of the rings in units of the corotation radius. Blue: giant planets; red: small bodies. Green dashed line: the corotation line. Brown dotted line: the 1:3 resonance. Short grey marks: the Roche limit for ice. Saturn's E ring, fed by Enceladus, is not shown. *How to read:* a ring to the left of the green line lies where material falls into the planet; a ring to the right lies where material is pushed outward.



Ring around a moon: none has been found around any moon. The suspected ring of Rhea was ruled out in 2010. Moons only feed the planet's rings (blue circles): Enceladus the E ring, Metis/Adrastea/Amalthea/Thebe the rings of Jupiter, Mab the μ ring of Uranus; Galatea holds Neptune's arcs; Phoebe feeds a huge ring off the figure (arrow in Saturn's row).

What it shows: at the giant planets the rings lie before the corotation line or around it, inside the Roche limit. At the small bodies (Chariklo, Haumea, Quaoar) the rings lie beyond the line. Mars: a ring in the past around the line, which has since migrated out to 6.03.

Figure A.2. Ring map: each body, in order from the Sun, against its corotation line (red dashed) and its Roche limit for ice (grey dotted). Distance in body radii, on a logarithmic scale. Gold: a ring visible in an amateur telescope; blue-grey: a faint ring; red: a ring of a small body; dotted frame: a ring in the past. A moon that feeds a ring is circled in blue. No ring has been found around any moon.

PART B Where the material comes from

B.1 Leakage from within

According to Paper 11, differentiation drives water and light material upward from the planet's interior (the pump principle), and the gas envelope closes over it (the cap principle). The cap closes, but it is not perfectly sealed. A little material leaks into the corotation zone, is trapped inside the Roche limit and feeds the ring. The ring is renewed for as long as differentiation continues and an outlet exists.

At Mars the feed passed through the volcano on the protrusion. While the ground was loose and let volatiles out, the ring was fed in large masses. As the spin slowed, sorting advanced and the crust cooled and thickened, the confinement became complete. The feed stopped and the ring dried [Paper 10].

B.2 Why it is not space dust

• **Purity:** the A and B rings are 95%–99% water ice [Cuzzi et al. 2010]. Interplanetary dust is a dark mixture of silicates, carbon and metals. A ring built from it would have been dark from the start.

• **The pollution acts from outside:** Cassini measured the rate at which micrometeoroids fall onto the rings [Kempf et al. 2023]. Space dust soils an existing ring; it is not what built it. The difference between the dark C ring and the bright A and B rings is explained the same way: at the same rate of pollution, a sparse ring darkens faster [Cuzzi & Estrada 1998].

• **Mass and shape:** 1.54×10^{19} kg lie in a disc tens of metres thick, in the equatorial plane. Dust arriving from every direction carries no common angular momentum and cannot settle into one such disc. A single source turning with the planet is needed.

• **The composition follows the cap:**

– **Saturn:** a thick cap keeps the water in, and the ring is almost pure ice.

– **Uranus and Neptune:** the cap is thin and the water escaped through the line [Paper 11, A.9]. The rings are narrow and dark.

– **Jupiter:** the cap is thickest, and the rings are thin dust fed by the inner moons [Owen et al. 1979].

– **Mars:** the ring was made of condensed vapours and crustal material [Paper 10].

B.3 A moon that feeds a ring

Saturn's E ring is fed by the jets of Enceladus. At Jupiter, the inner moons Metis, Adrastea, Amalthea and Thebe release the dust of the rings. The direction is always the same: a moon feeds a ring, and a ring does not feed a moon.

B.4 Two ways, and when a ring lives

All ring material comes from the parent, in two ways:

- **Directly:** material ejected from the ground that did not cross the corotation line. This is lava, water, vapour and crustal dust. It orbits in the ring, is ground and disperses, and does not become a moon. This is how Saturn's D, C and B rings are fed, though they contain no moons, and how the Martian ring was fed [Paper 10].
- **Through moons:** moons that came out of the parent and go on releasing material.

– Enceladus feeds the E ring with water jets.

– Metis, Adrastea, Amalthea and Thebe feed all of Jupiter's rings with dust knocked off by impacts [Burns et al. 1999].

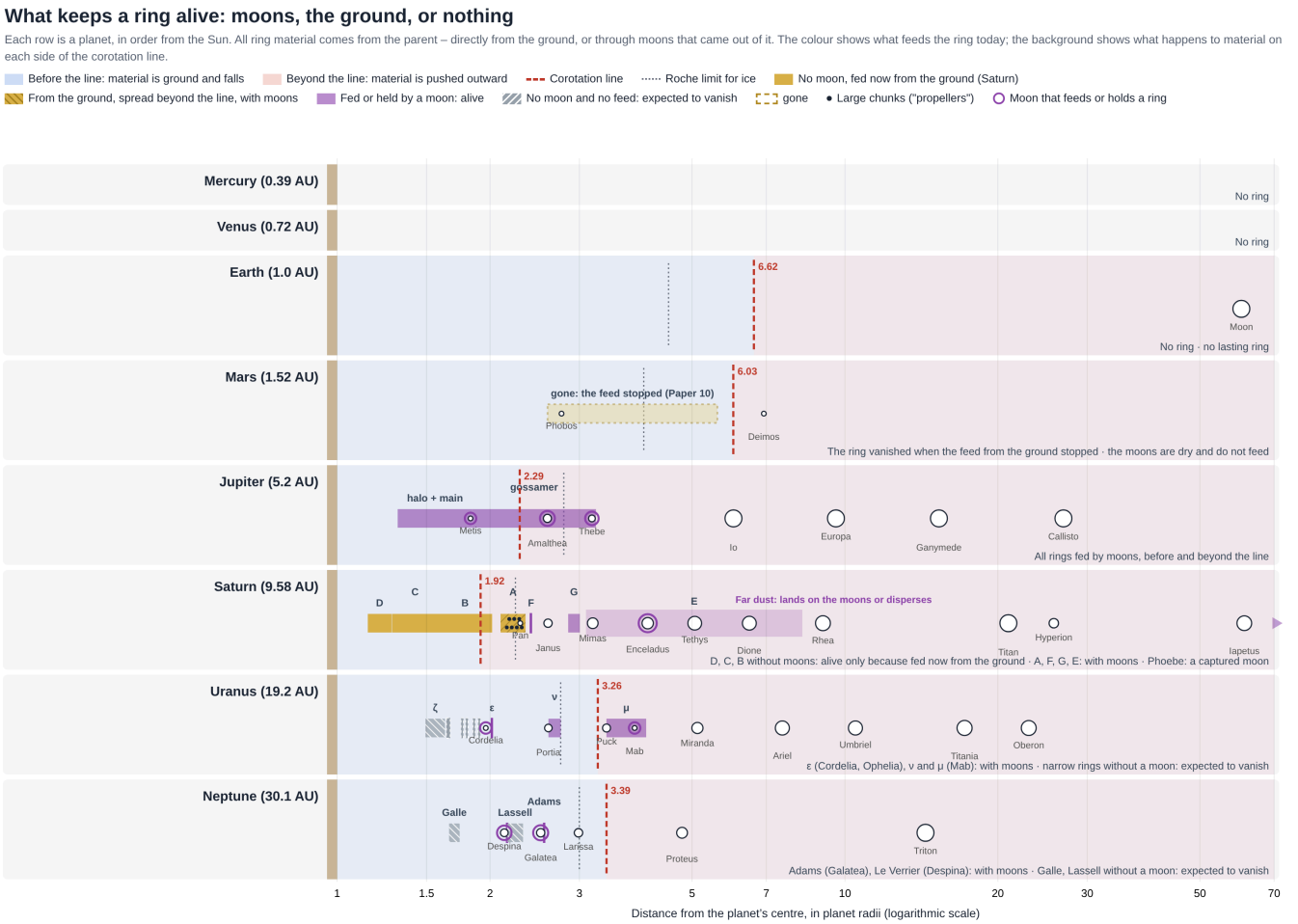
– Mab feeds the μ ring of Uranus.

A ring lives as long as one of the two ways feeds it:

- A moon feeds a ring only if it releases material. Dry moons, such as Phobos and Deimos, do not feed.
- When the feed from the ground stops and the moons are dry, the ring vanishes. This is how the Martian ring vanished.
- A ring with no moon and no feed from the ground has passed the stage of moons and is expected to vanish (F.3).

Far material disperses into space or lands on the moons. The charged gas turning with the planet pushes dust beyond the line outward. Far from the planet, sunlight slows the dust and it drifts inward. Dust from the Phoebe ring lands on Iapetus and forms its dark side [Verbiscer et al. 2009]. Dust from the E ring coats Mimas, Tethys, Dione and Rhea [Verbiscer et al. 2007].

A captured moon is not part of the rule: it did not come from the parent, and the rule for identifying it was set in Paper 11. The Phoebe ring, from a captured moon, is outside its scope.



The rule: all ring material comes from the parent, in two ways: directly – material ejected from the ground (lava, water, vapour) that did not cross the line; it orbits in the ring, is ground and disperses, and does not become a moon – or through moons that came out of the parent and go on releasing material. A ring lives as long as one of the two ways feeds it. When the feed from the ground stops and the moons are dry, the ring vanishes (Mars). A ring with no moon and no feed from the ground is expected to vanish. Far material disperses into space or lands on the moons.
Outside the rule: a captured moon (Phoebe) did not come from the parent; it is defined in Paper 11.

Figure B.1. What keeps a ring alive. Each row is a planet, and the colour shows what feeds the ring today:
– Gold: fed from the ground, with no moon (Saturn).
– Gold hatched: from the ground, spread beyond the line, with moons and large chunks (the A ring).
– Purple: fed or held by a moon.
– Grey hatched: no moon and no feed, expected to vanish.
– Dotted frame: gone.
Background: blue before the corotation line, where material is ground and falls; pink beyond it, where material is pushed outward.

B.5 Two conditions for a massive ring

Not every planet with moons has a massive ring. Two simple questions decide whether a planet will have a massive ring fed from the ground. Think of a net that catches balls: the net must be where the balls arrive, and someone must throw them.

Condition 1 – the net in the right place: does the corotation line lie before the Roche limit? Material leaking from the ground arrives at the corotation zone (B.1). If the line lies before the Roche limit, the material enters the zone where the tidal force keeps it from

gathering, and it stays a ring. If the line lies beyond the Roche limit, material reaching it is no longer caught.

The position of the line and the Roche limit both depend on the planet's density. When one is divided by the other, the density cancels:

$$\frac{x_c}{x_R} = \left(\frac{P}{P_*}\right)^{2/3}, \qquad P_* = 2\pi\sqrt{\frac{3 \cdot 2.46^3}{4\pi G \rho_{ice}}} \approx 13.4 \text{ h}$$

Here P is the planet's day and $\rho_{ice} = 900$ kg m⁻³. This gives the simple form of the condition: **the line lies before the Roche limit if and only if the day is shorter than about 13.4 hours**. This holds for any planet, of any size and density.

Spin is therefore the condition itself, and gravity does not change it: Jupiter has the strongest gravity, and Saturn a gravity almost equal to the Earth's.

Condition 2 – someone throwing: is there leakage from the ground? The gas cap does not close everything, and the outer layer has neither frozen nor dried.

Planet and time	Day (h)	Line	Roche limit	Condition 1	Condition 2	Result
Saturn today	10.6	1.92	2.24	Yes	Yes: the cap leaks	**Massive ring from the ground**
Young Mars	7–9	1.74–2.05	2.68	Yes	Yes: the volcano on the protrusion	**It had a ring**
Jupiter today	9.9	2.29	2.80	Yes	Almost none: the thickest cap	Dust from moons only
Earth at birth	5.8	2.29	4.00	Yes	No: it dried fast	No ring; the Moon was built
Uranus today	17.2	3.26	2.76	No	No: the water escaped, frozen	Narrow remnants and moon rings
Neptune today	16.1	3.39	3.00	No	No: frozen	Narrow remnants and moon rings
Mars today	24.6	6.03	4.02	No	No: the ground closed	The ring vanished
Earth today	23.9	6.62	4.49	No	No	No ring

Distances are in planet radii. For young Mars and the Earth at birth they are in the radii of that time, when the planet was expanded. The Roche limit for ice was computed as $2.46\,(\rho/900)^{1/3}$, with the planet's density at that time.

What it shows: only when both conditions hold together is there a massive ring from the ground – Saturn today and young Mars. Each of the other cases fails at least one condition:

- **Jupiter** spins fast, but its thick cap hardly leaks.
- **The Earth at birth** spun fast, but it had dried.
- **Uranus and Neptune** turn too slowly, with days of 16–17 hours, so their lines lie beyond the Roche limit. They have also frozen, and only narrow rings are left to them.

On Mars the two conditions are seen to vanish one after the other. As the spin slowed, the line passed the Roche limit and the feed from the ground stopped. The ring then vanished.

This also accounts for distance from the Sun. Like the arc of moon births [Paper 11], rings peak at Saturn. Near the Sun planets dry out, and far from it they freeze; in both cases there is no leakage.

Two conditions for a massive ring

A net that catches balls: the net (the Roche limit) must be where the balls arrive (the corotation line), and someone must throw balls (leakage from the ground).

Capture zone: before the Roche limit | Roche limit for ice | Corotation line (young Mars: a range)

Planet and time	Where the line is, and where the Roche limit is (in planet radii)	Condition 1: day shorter than 13.4 h (line before the Roche limit)?	Condition 2: leakage from the ground?	Result
Jupiter today	day 9.9 h · line 2.29 · Roche 2.8	Yes	Almost none: the thickest cap	Dust from moons only
Saturn today	day 10.6 h · line 1.92 · Roche 2.24	Yes	Yes: the cap leaks	**Massive ring from the ground**
Uranus today	day 17.2 h · line 3.26 · Roche 2.76	No	No: the water escaped, frozen	Narrow remnants and moon rings
Neptune today	day 16.1 h · line 3.39 · Roche 3	No	No: frozen	Narrow remnants and moon rings
Young Mars	day 7–9 h · line 1.74–2.05 · Roche 2.68	Yes	Yes: the volcano on the protrusion	**It had a ring**
Mars today	day 24.6 h · line 6.03 · Roche 4.02	No	No: the ground closed	The ring vanished
Earth at birth	day 5.8 h · line 2.29 · Roche 4	Yes	No: it dried fast	No ring; the Moon was built
Earth today	day 23.9 h · line 6.62 · Roche 4.49	No	No	No ring

Only when both conditions hold together is there a massive ring from the ground: Saturn today and young Mars. On Mars, as the spin slowed, the line passed the Roche limit and the feed stopped – and the ring vanished. For young Mars and the Earth at birth, distances are in the radii of that time, when the planet was expanded.

Figure B.2. Two conditions for a massive ring. Condition 1 is equivalent to a day shorter than 13.4 hours. Each row shows the capture zone (blue) up to the Roche limit

(dotted line), and the position of the corotation line (red). Green: the condition holds; pink: it does not. *How to read:* if the red line lies inside the blue zone, condition 1 holds.

PART C A ring breaks and grinds; it does not build

C.1 Inside the Roche limit

Inside the Roche limit, the tidal force exceeds self-gravity, so particles cannot gather into one body. A large chunk that enters the zone is torn apart and ground down in collisions to gravel and dust. Saturn's A ring ends exactly at the Roche limit for ice, and no moon has gathered from it beyond that limit.

C.2 Outside the Roche limit as well

Quaoar's outer ring lies at 4,057 km, 2.4 times the Roche limit for ice. On the accepted understanding, material at that distance should gather into a moon within decades. Yet the ring is dense and present [Morgado et al. 2023]. In the literature this is a puzzle. In the mechanism it is the expected result: a ring does not build moons, even where the Roche limit does not prevent it. Chariklo's rings also lie slightly outside the Roche limit.

C.3 The ring moons

The tiny moons inside and at the edges of Saturn's rings have densities of only 340–640 kg m^{-3} [Porco et al. 2007]. They are Pan, Daphnis, Atlas, Prometheus, Pandora, Janus and Epimetheus. According to Paper 11, they are remnants of moons that were ground and polished in the ring, and ring material then accumulated on them. Accumulation on an existing body is not the building of a moon. The model of Crida and Charnoz [2012] has moons gathering from a ring that spreads beyond the Roche limit; it rests on calculation, not on observation.

C.4 A ring as a site of destruction

Wisdom and colleagues [2022] proposed that Saturn's rings formed from an ancient moon that approached, crossed the Roche limit and broke apart. This proposal, too, treats the ring as a site of break-up, not of building.

PART D Dynamics: what returns and what leaves

D.1 Most material returns inward

In a ring, near particles orbit faster than far ones. The collisions between them turn orbital energy into heat while the total angular momentum is conserved. To satisfy both conditions, most of the mass moves inward and a little is pushed outward, carrying the angular momentum with it [Lynden-Bell & Pringle 1974]. The ring is therefore a funnel: most of its material returns to the planet.

D.2 The line divides

Whatever turns with the planet brakes material below the corotation line and drops it inward, and pushes material above the line outward. On moons and large chunks it is the tide that acts; on charged dust it is the magnetic field, which turns with the planet. A dense, massive ring behaves mainly as one spreading body (D.1). A ring also transfers momentum to the moons at its edges and pushes them outward [Goldreich & Tremaine 1980].

The Martian ring worked in both ways [Paper 10]:

- It pushed Deimos outward.
- Its gaseous inner edge held Phobos on the corotation line and led it outward with the line, until the ring dried near 19,000 km.

D.3 Saturn: a steady flow

Cassini measured the material falling from the rings into Saturn in two channels:

- **Neutral material** flowing into the equator: 4,800–45,000 kg s^{-1} [Waite et al. 2018].
- **Ionised material** flowing along the magnetic field lines: 432–2,870 kg s^{-1} [O'Donoghue et al. 2019].

Infall rate (kg s^{-1})	Time to empty the ring without feeding	Feed required over 4.5 billion years	As a fraction of Saturn's mass
432	1.1 billion years	4 ring masses	1×10^{-7}
2,870	170 million years	26 ring masses	7×10^{-7}
4,800	100 million years	44 ring masses	1.2×10^{-6}
45,000	11 million years	415 ring masses	1.1×10^{-5}

Without feeding, the ring would vanish in far less time than the age of the Solar System. The literature therefore concludes that the rings are young [Iess et al. 2019; Kempf et al. 2023].

In the mechanism, the ring is a steady flow in balance. It is renewed from below and drains as rain. It looks young because its material is constantly replaced, though the process is old. The steady-state requirement is that the rate of leakage from the parent into the ring equals the rate of ring rain. The feed needed over Saturn's whole life is less than one part in a hundred thousand of its mass, so it is within reach.

D.4 Where the angular momentum went: moons or water

When a planet's corotation line moves outward, the planet has lost angular momentum. The question is what took it: the moons, or water that left through the line. The calculation compares the momentum lost with the orbital momentum of today's regular moons. Moment-of-inertia factor: about 0.22 for Uranus and about 0.23 for Neptune.

Planet	Change of day	Momentum lost	Momentum of the moons	Moons' share	Water needed through the line
Uranus	8.3 → 17.2 h	1.3×10^{36}	1.4×10^{34}	1%	2.5×10^{24} kg, about 2.9% of the mass
Neptune	7.3 → 16.1 h	1.9×10^{36}	5×10^{31}	0.003%	3.2×10^{24} kg, about 3.1% of the mass
Jupiter	9.8 → 9.9 h	The moons carry about 1% of the spin momentum		All of it	None needed
Saturn	10.4 → 10.6 h	The moons carry about 1.4% of the spin momentum		All of it	None needed

Momentum in kg m² s⁻¹. The water mass is computed for water leaving from the line of that time, at 2 radii [Paper 11].

- **Uranus and Neptune:** the moons are too small; the water took the momentum. The water did not reorganise into moons. All of Uranus's regular moons together weigh 9×10^{21} kg, while the water that left weighed 280 times as much. Had the water gathered into moons, the moons would hold the momentum. The water was therefore expelled outward, into space.
- **Jupiter and Saturn:** the line has hardly moved, and the moons alone account for the small momentum transferred. No escaping water is needed, consistent with the thick cap. The rings carry no significant momentum either: Saturn's rings weigh about one ten-thousandth of its moons.

- **A moon that takes its water back:** most of the ice grains in the jets of Enceladus fall back on its surface as snow, and only a small part escapes to the E ring [Kempf et al. 2010].

Why the far planets lost water

One might expect the planets near the Sun to lose water and the far, cold ones to keep it. Among the giant planets the opposite happens, for three reasons:

1. The cap. Distance from the Sun set how much hydrogen and helium each planet gathered before the disc dispersed [Paper 11]. Far from the Sun solid material builds up slowly. By the time the inner body is large enough to draw in gas, most of the gas is gone. Jupiter's cap (90%–97% gas) and Saturn's (70%–85%) are therefore thick, and those of Uranus and Neptune (10%–20%) thin. A thick cap holds the water in; a thin cap lets it out.

2. Escape speed. The escape speed is a threshold: the least speed material needs to leave the planet and not fall back. A low escape speed does not make escape harder; it makes it easier. Picture climbing out of a pit. Jupiter is a deep pit: it takes a jump of 59.5 km/s to get out. Saturn is a pit of 35.5. Uranus and Neptune are shallow pits, of 21.3 and 23.5. The same water, given the same push by the spin, gets out of the shallow pit and stays in the deep one.

3. The exit is mechanical, not thermal. The water did not evaporate. It reached the corotation line while the planet was spinning fast, with a day of 7–8 hours. Beyond the line it was pushed outward. Here too, spin decides, not temperature.

Cold does not break water down. Water is broken mainly by the Sun's ultraviolet light, which is strong close to the Sun; this is how Venus lost its water. Far from the Sun water freezes into ice. Near the Sun planets dry out from heat and light, whereas among the giant planets the cap decides. These are two different mechanisms.

The size of the moons relative to the planet

System	Mass of all regular moons, as a fraction of the planet's mass
Earth	1.2%
Jupiter	0.021%
Saturn	0.025%
Uranus	0.010%
Neptune (without Triton, captured)	0.00005%

Mars	0.000002%

PART E Sorting in the ring

E.1 Sorting by collision, not by settling

Inside a planet, dense material sinks and light material floats. In a ring this does not happen: a dense particle and a light one at the same distance orbit at the same speed. Sorting in a ring is done by collisions, because different materials have different power to move one another. The image is a road that stays clean because the cars throw the small particles to the sides. A heavy, hard body keeps its course; a light or soft particle takes the kick, is thrown aside and ground down.

E.2 What has been measured

• **"Propellers":** in the A ring Cassini imaged dozens of propeller-shaped structures [Tiscareno et al. 2006]. At the centre of each lies a body about a hundred metres across, with lanes emptied of material on either side, several kilometres long. This is the road observed directly: a hard body clears a lane for itself and pushes the light particles aside. The propellers were found in the A ring, beyond the corotation line. There material does not fall into the planet, and chunks survive.

• **Thickness:** in the dense rings the chunks, centimetres to metres across, lie in a layer about 10 m thick. Small particles gain larger random speeds in collisions and spread into a thicker layer.

• **Size distribution:** the number of chunks falls with size as a power law, about $n(D) \propto D^{-3}$ [Zebker et al. 1985], as expected from a mill that crushes chunks.

• **Uranus:** narrow rings of chunks centimetres to metres across, with bands of fine dust between them.

E.3 The element of time

The size of a chunk tells its age in the ring: a large chunk is new, because it has not yet been ground. Chunks the size of small moons enter the ring, break up in its middle part and are ground to dust. By D.1, most of that dust falls into the planet and a little is pushed outward. The ring is therefore a conveyor: a large chunk goes in and dust comes out. This gives a prediction: the large bodies in a ring will be concentrated where the ring is fed.

PART F The life cycle of a ring

F.1 Migration of the line erases rings

The position of the corotation line depends on the planet's spin. The planet can slow in two ways: through the tide of a large moon, or through water that escapes through the line and carries angular momentum. When it slows, the line migrates outward. Material that lay on the line or above it now lies below it; the tide brakes it and it falls. If the feed stops at the same time, the ring drains and disappears.

F.2 The stages

1. Feeding: the line is close, differentiation is active, and material leaks out and is trapped. The ring is massive (Saturn today; young Mars).

2. Decline: sorting advances, the crust thickens or the cap leaks, the feed falls and the line migrates outward. Narrow rings remain below the line (Uranus and Neptune).

3. Disappearance: the feed has stopped and the ring has fallen. What remains is the moon that crossed the line and the material that fell on the planet (Mars: the dust, and Phobos falling [Paper 10]).

On the Earth no lasting ring could exist: the pear that built the Moon swept the equatorial band in which a ring would have formed and gathered its material into the Moon [Paper 10].

F.3 Rings expected to vanish

By the rule of B.4, a ring with no moon and no feed from the ground is expected to vanish:

Planet	Rings with a moon	Rings without a moon	Reading
Saturn	A (Pan, Daphnis), F (Prometheus, Pandora), G (Aegaeon), E (Enceladus)	D, C, B	D, C and B live only because they are fed now from the ground. The measured ring rain (D.3) shows that without feeding they would vanish.
Uranus	ε (Cordelia, Ophelia), ν (between Portia and Rosalind), μ (Mab)	ζ, 6, 5, 4, α, β, η, γ, δ	Expected to vanish. In the literature, what holds them is an open question; small moons to hold them have not been found.
Neptune	Adams (Galatea), Le Verrier (Despina)	Galle, Lassell	Expected to vanish

Jupiter	All	—	Live as long as the inner moons release dust
Mars	—	The ring of the past	Vanished when the feed from the ground stopped; the moons are dry

PART G The mark of the ring on the moon

G.1 The rule

A fully smooth silicate moon shows that it stayed in a ring.

Deimos is the smoothest small body observed near Mars. Shape models by Ernst et al. [2023] were measured the same way for both moons. They show that the median depth of the hollows on Deimos is 1.9% of their width, against 5.5% on Phobos. 84% of the hollows on Deimos are shallower than 6% of their width, against 56% on Phobos [Paper 10, J.1]. Its orbit is almost a perfect circle (eccentricity 0.0003).

A circular, equatorial orbit together with a smooth face marks a body that stayed long in the heart of a ring. The other two bodies fit the same reading:

- **Phobos** left through the lower edge of the fading ring, so it was ground only slightly and kept its grooves.
- **The Earth's Moon:** the Earth had no lasting ring, and the Moon keeps a saturated crater record.

Among the planets, Deimos is the only fully smooth silicate moon, and its planet has no ring today. The smooth icy moons (Europa, Enceladus and Triton) are geologically active and are not covered by the rule.

G.2 The material decides

An ice ring adds; a silicate ring grinds.

- **Ice:** cold ice grains are coated with frost and stick to one another in slow contact [Bridges et al. 1984]. This is how the equatorial ridges of Pan, Atlas and Daphnis were built. The ridges stand above the body itself and give it a "saucer" shape [Charnoz et al. 2007].
- **Silicate:** the Martian ring was made of condensed vapours and silicate crustal material. Silicate grains are hard and act like sandpaper, so Deimos was ground rather than built up.

G.3 Gravity

A large moon holds with its gravity the material that falls on its equator. Among small moons gravity does not decide, and the kind of material does:

Moon	Diameter (km)	Density (kg m⁻³)	Surface gravity (m s⁻²)	Ring	Mark
Methone	2.9	310	0.0001	Ice (arc near the G ring)	Smooth, coated
Daphnis	7.6	340	0.0004	Ice (A)	Equatorial ridge
Pan	28	400	0.0016	Ice (A)	Equatorial ridge
Atlas	30	460	0.0019	Ice (edge of A)	Equatorial ridge
Deimos	12.4	1,471	0.0025	Silicate and vapour	Ground all over
Phobos	22.2	1,860	0.0058	Fading gaseous edge	Ground slightly
Iapetus	1,469	1,088	0.22	?	Equatorial ridge (G.6)

Deimos has a stronger gravity than Pan and Atlas, yet material built up on them and not on Deimos. The difference lies in the material.

G.4 Ring thickness

The moon and the ring lie in the same plane, so the contact is concentrated on the moon's equator. The ratio of the moon's diameter D to the ring's thickness H sets the shape of the mark:

Ratio	What happens	Mark	Example
$D < H$	The moon is immersed in the ring on all sides	Full smoothing	Deimos, Methone
$D > H$	The poles stand above and below the ring	A broad equatorial belt	Pan, Atlas, Daphnis
$D \gg H$	The ring is a thin thread around a large sphere	A narrow ridge, or a mark since erased	Iapetus (?)

Deimos left at a diameter of 14–22 km [Paper 10, Table 18]. For it to be smoothed all over, the ring had to be at least tens of kilometres thick. A ring of condensed vapour and dust, like the Martian one, is far thicker than a ring of ice chunks, and so meets this requirement.

G.5 Crossing speed

The rate of tidal migration of a moon grows with its mass:

- **A small moon** recedes slowly, stays long in the ring and carries a strong mark.
- **A large moon** is pushed out quickly and crosses the ring in a short time. It carries a shallow mark that is easily erased.

The speed of the encounter matters as well. A moon that moves with the ring particles receives gentle contact. A moon that the ring sweeps past at speed is ground.

G.6 Iapetus: an open question

Iapetus has a ridge on its equator, about 20 km wide, up to 20 km high and about 1,300 km long. Two explanations are possible:

(a) The ridge was carved when Iapetus crossed Saturn's ring near its birth site, at 2.53 radii [Paper 11].

(b) The ridge formed from a sub-ring that circled Iapetus itself and collapsed onto it [Ip 2006; Levison et al. 2011; Dombard et al. 2012].

The second contradicts the rule that moons have no rings (J.1), but it is established in the literature. The question remains open (Part M).

PART H Signatures on the parent planet

H.1 Saturn: measured

In its final dives in 2017, Cassini flew between the planet and the inner ring and measured the falling material:

- **Composition:** water, methane, ammonia, carbon monoxide, nitrogen and hydrocarbons flowing from the ring into the equator [Waite et al. 2018], and nanometre silicate grains [Hsu et al. 2018].

- **Bands in the ionosphere:** dark infrared bands at latitudes 25°–55°. Charged water molecules flow from the ring along the magnetic field lines and neutralise the H_3^+ ions [O'Donoghue et al. 2013].

The ring therefore leaves two signatures on its parent. Neutral material falls on the equator. Charged material falls at middle latitudes, on both sides of the equator.

H.2 Mars: required

When the Martian ring dried, its material fell on Mars and is the source of the Martian dust [Paper 10]. That material was vapour and some crustal material ground to dust. The search directions follow:

- **Equatorial deposits with no source of flow:** the Medusae Fossae Formation is a low-density, thick deposit along the equator. The accepted explanation is volcanic ash [Ojha & Lewis 2018].
- **Isotopes:** hydrogen with a D/H ratio different from its surroundings and similar to that of Phobos and Deimos.
- **Igneous layers at Mount Sharp:** ChemCam identified dozens of igneous rocks in Gale crater [Cousin et al. 2017], and a dark, resistant marker layer runs across the mountain.
- **Bands at middle latitudes:** if Mars had a magnetic field while the ring existed, the charged material would have landed in symmetric bands on both sides of the equator, as at Saturn.

H.3 A test that did not pass: Atacama

On 25 April 2026, Curiosity drilled into a flat block named Atacama, about 45 cm across and 13 kg, that was not attached to the ground. The block stuck to the drill and was freed only six days later [NASA/JPL 2026]. Such detached blocks are scattered across the area, in Gale crater, about 320 km from the equator, and the possibility that they are ring remnants was examined.

In fact, the block belongs to the layered sulfate unit of Mount Sharp. It is a finely layered sediment, rich in hydrated sulfate salts, with the same composition as the layers beneath it. It is not a ring remnant. The test defines the criteria for a true remnant:

Measure	What was found at Atacama	What a ring remnant requires

Type	A layered sediment evaporated from water	Volcanic or spongy material, with no water layering
Minerals	Hydrated sulfates	Dry minerals, or minerals that were heated
Exterior	Desiccation cracks	A fusion crust from atmospheric entry
Relation to the surroundings	Same composition as the layer beneath	Composition foreign to the local layer

PART I The ring and the Sun

• **"Spokes":** dark bands appear across the B ring, turn with it and vanish after hours [Smith et al. 1981]. They are seasonal and multiply around equinox, when the Sun lights the ring at a shallow angle [Mitchell et al. 2006]. Tiny grains become electrically charged and levitate above the ring plane.

• **Ionised rain:** the Sun's ultraviolet light ionises water molecules in the ring; they are trapped on field lines and fall into the planet (H.1). The Sun thus speeds the draining of the ring.

• **Radiation drag:** sunlight slows tiny dust grains, and they drift inward (the Poynting–Robertson effect).

PART J Moons and small bodies

J.1 Moons have no rings

No ring has been found around any moon. The suspected ring around Rhea was tested and ruled out [Tiscareno et al. 2010]. The planet's gravity rules the region, so material released from a moon joins the planet's ring (B.3).

J.2 Small bodies with rings

Three small bodies have dense rings: Chariklo [Braga-Ribas et al. 2014], Haumea [Ortiz et al. 2017] and Quaoar [Morgado et al. 2023; Pereira et al. 2023]. Unlike the rings of the giant planets, all lie above the corotation line (Table A.3). Four of the five rings lie between 2.01 and 2.18 corotation radii, close to the 1:3 resonance at $3^{2/3} a_c = 2.08\, a_c$.

Haumea turns in 3.9 hours, and the ends of its long axis (1,161 km) reach beyond its corotation line (1,104 km) [Paper 11, K.2]. Material released from the ends is above the line; it

is pushed outward and stops at the resonance. In the mechanism, Haumea's ring is material leaving a body on the threshold of birth, and it does not gather into a moon.

Quaoar turns more slowly, and its surface lies far from the line (3.6 radii), so the source of its rings remains open. But its outer ring is the strongest evidence that a ring does not build (C.2): it lies far outside the Roche limit and is still not a moon.

PART K Review by body

Body	State today	Source of material	Position relative to the line	Reading in the mechanism
Earth	None	—	—	The pear swept the equatorial band; no lasting ring
Mars	None; had one	Directly from the ground: vapour and crustal material, through the volcano on the protrusion	On the line, which migrated outward	Fed, pushed Deimos, led Phobos; vanished when the feed stopped, 3.7–3.5 billion years ago; the moons are dry and do not feed [Paper 10]
Jupiter	Thin dust	Through moons: Metis, Adrastea, Amalthea, Thebe	Around the line	The thickest cap; almost no leakage. Alive as long as the moons release dust
Saturn	Massive, pure ice	Directly from the ground (D, C, B, A); through moons (F, G, E)	Around the line; A beyond the line, its edge at the Roche limit	Steady flow: leakage equals rain. A is the parent's ring spread beyond the line
Uranus	Narrow and dark	ε, ν, μ: moons; the rest: no moon	Below the line (0.51–0.62); μ beyond it	Thin cap, the water escaped, the line moved to 3.24. Rings without a moon are expected to vanish
Neptune	Narrow, with arcs	Adams, Le Verrier: moons; Galle, Lassell: no moon	Below the line (0.50–0.75)	As Uranus
Haumea	Dense ring	Ends of the body crossing the line	Above the line (2.07)	A body on the threshold of birth; the material stops at the 1:3 resonance
Chariklo	Two rings	Open	Above the line (2.10–2.18)	At the 1:3 resonance

Quaoar	Two rings	Open	Above the line (1.25; 2.01)	Outside the Roche limit and not a moon

PART L Requirements, predictions and refutation criteria

Requirements:

1. At Saturn, the rate of leakage from the parent into the ring equals the rate of ring rain.

2. Ring material matches in its isotopes the outer layer of the parent, not comets or interplanetary dust.

3. No moon has gathered from ring material.

4. A fully smooth silicate moon stayed in a ring.

Predictions:

1. MMX (from 2027) will find old, shallow, worn hollows on Deimos and the same composition as Phobos. On Phobos it will find only slight grinding.

2. Small, elongated, fast-spinning bodies whose ends lie near the corotation line will hold rings close to $2.08\,a_c$.

3. In Saturn's rings, the large bodies will be concentrated where the ring is fed (E.3).

5. Uranus's narrow rings without a moon, and Neptune's Galle and Lassell rings, will be found to have no moon feeding them. A future mission will find them thinning.

4. On Mars, equatorial deposits of ring material will be found, with a composition different from the local layer and similar to that of the moons.

Refutation criteria:

1. A direct observation of a moon gathering from ring material.

2. A fully smooth silicate moon that never stayed near a ring.

3. A ring whose isotopic composition differs from that of its parent and matches an external source.

4. A massive ring lying wholly far above the corotation line of a large planet, with no resonance holding it.

5. A ring with no moon and no feed from the ground that persists over time without thinning.

PART M Open questions

1. Iapetus: did the ridge form by crossing Saturn's ring, or from a sub-ring (G.6)?

2. Quaoar and Chariklo: what is the source of the material in the rings of bodies whose surfaces lie far from the corotation line?

3. Jupiter: is there leakage from within in addition to dust from the moons, or does the thick cap close completely?

4. Mars: where did the ring material fall, and how thick was the ring?

5. Saturn: a direct measurement of the rate of leakage into the ring, to compare with the rate of ring rain.

DATA & CODE Data and code

A short Python script computed the following from basic formulas and the cited data:

- the positions of the corotation line and the Roche limit;
- the surface gravity of the moons;
- the emptying and feeding times of Saturn's rings;
- Figure A.1.

The script will be added to the series' code package.

- **Repository:** Zenodo
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DECLARATIONS Declarations

Use of artificial intelligence. The author used Claude (Anthropic) for calculations, for gathering data from the sources, for building the figure and for drafting. He also used a conversation with Gemini (Google) to examine ideas and data. The ideas and claims of the model are the author's. All data were checked against the cited sources.

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